



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2009

**YEAR 12 Mathematics Extension 1
HSC Task #2**

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- All answers must be given in exact simplified form unless otherwise stated.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 61

- Attempt questions 1-3
- Start each new section of a separate answer booklet

Examiner: *D.McQuillan*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

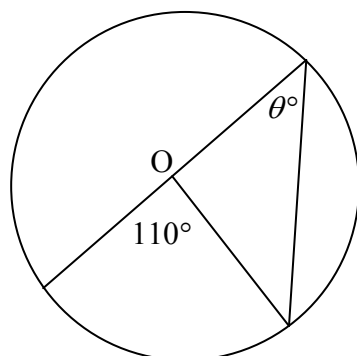
NOTE: $\ln x = \log_e x, \quad x > 0$

Section A (21 marks)

Marks

- (1) O is the centre of the circle. Find the value of θ .

1



- (2) Find

5

(a) $\int 3x^5 dx$

(b) $\int \frac{(3x+2)^2}{3} dx$

(c) $\int \frac{dx}{\sqrt{7x-1}}$

- (3) Evaluate

5

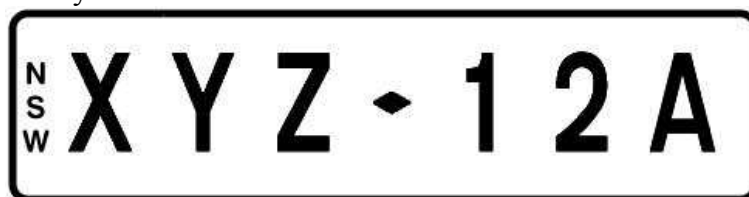
(a) $\int_{-8}^8 \frac{x^3}{2} dx$

(b) $\int_{-3}^3 (x^2 + 3) dx$

(c) $\int_0^4 \sqrt{4-x} dx$

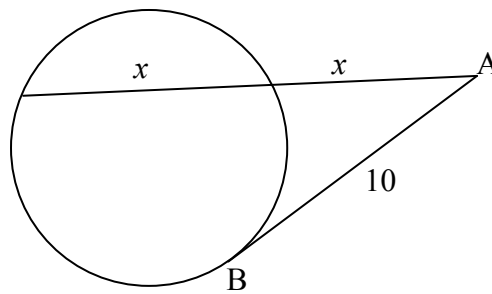
- (4) The new NSW number plates have three letters then two numbers followed by a letter.

1



How many number plates with this configuration are possible?

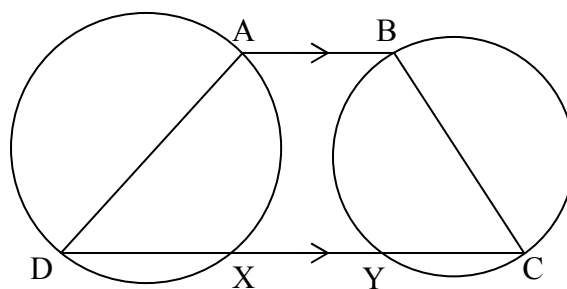
- (5) AB is a tangent. Find the value of x . 2



- (6) Given the parametric equations $x = 2t - 1$ and $y = 4t^2$ find $\frac{dy}{dx}$. 2

- (7) Find the volume when $y = x^2 - 3$ is rotated about the y-axis between $y = -3$ and $y = 2$. 2

- (8) ABCD is a trapezium in which $AB \parallel DC$. AD and BC are the diameters of their respective circles, and these circles cut DC at X and Y respectively. Prove that ABYX is a rectangle. 3



End of Section A

Start a new ANSWER BOOKLET

Section B (20 marks)

Marks

- (1) Use Simpson's Rule with 3 function values to find an approximation to the area under the arc of $y = \log_{10} x$ between $x = 2$ and $x = 4$. Round answer to 2 decimal places. 2
- (2) For the function $f(x) = x^3 - 3x + 1$. 5
- (a) Show that $f(x)$ has a zero for x between 0 and 1.
- (b) Take $x = 0$ as your first approximation and use two applications of Newton's method to find a better approximation to 3 decimal places.
- (c) Explain why you could not use $x = 1$ as your first approximation.
- (3) A five-card hand is dealt from a standard 52-card deck. How many ways can you be dealt, 3
- (a) three hearts and two clubs.
- (b) three of one suit and two of another.
- (4) Find the area enclosed between $y = x + 2$ and $y = x^2 - 2x + 2$. 3
- (5) If ${}^nP_k = 120 {}^nC_k$, find the value of k . 2
- (6) Given that the equation for the chord of contact is $xx_0 = 2a(y + y_0)$. Find the point of intersection of the tangents to $x^2 = 8y$ at the points where the chord of contact $y = x - 1$ intersects the parabola. 2

- (7) For the parabola $x^2 = 4ay$. 3
- (a) Derive the equation of the chord joining the points $P = (2ap, ap^2)$ and $Q = (2aq, aq^2)$.
- (b) Show that as q approaches p the equation of the chord becomes the equation of the tangent.

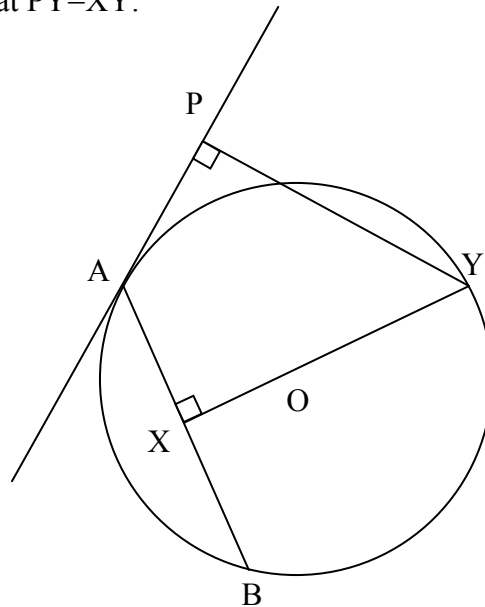
End of Section B

Start a new ANSWER BOOKLET

Section C (20 marks)

Marks

- (1) Prove that ${}^nP_k = {}^{n-1}P_k + k \cdot {}^{n-1}P_{k-1}$. 2
- (2) AB is a chord of a circle and PA is a tangent at A. XY is a perpendicular bisector of AB and passes through the centre of the circle, O. PY is perpendicular to PA. 6
- (a) Copy the diagram into your answer booklet.
- (b) Show that triangle AYB is an isosceles triangle.
- (c) If $\angle PAY = \alpha$ show that $\angle YAB = \alpha$.
- (d) Prove that $PY = XY$.



- (3) In how many ways can the letters of the word TOMATO be arranged if the Ts are to be separated? 2
- (4) A Rugby League team of 13 players are to be selected at random from 20 boys. What is the probability that of 3 friends, 2 are selected and one misses out. 2

(5) For the parabola $x = 4t$, $y = 2t^2$. 4

(a) Derive the equations of the normals at $t = p$ and $t = -p$.

(b) The normals intersect at R . Find the locus of R .

(6) Use mathematical induction to prove that 4

$$\frac{1}{2} + \frac{2}{3} + \cdots + \frac{n}{n+1} < \frac{n^2}{n+1}$$

for all integers $n \geq 2$.

End of Section C

End of Exam

SECTION A

1/ $\theta = 55^\circ$ 1

2/ (a) $\int 3x^5 dx = \frac{x^6}{2} + C$ 1

(b) $\int \frac{(3x+2)^2}{3} dx = \frac{(3x+2)^3}{27} + C$ 2

or $= x^3 + 2x^2 + \frac{4}{3}x + C$

(c) $\int \frac{dx}{(7x-1)^{1/2}} = \frac{2}{7} \sqrt{7x-1} + C$ 2

3/ (a) $\int_{-8}^8 \frac{x^3}{2} dx = \left[\frac{x^4}{8} \right]_{-8}^8 = 10$ 2

[Could use $\int_{-a}^a f(x) dx = 0$ if $f(x)$ odd]

(b) $\int_{-3}^3 (x^2+3) dx = \left[\frac{x^3}{3} + 3x \right]_{-3}^3 = 36$ 2

[Could use $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ if $f(x)$ even]

(c) $\int \sqrt{x-4} dx = \frac{2}{3} (x-4)^{3/2} + C$ 1

4/ $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 26 = 45697600$ ✓

5/ $(AB)^2 = 2x \times x$ 1

$100 = 2x^2$

$x = \sqrt{50} = 5\sqrt{2}$ 1

6/ $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$

$= 8t \div 2$ 2

$= 4t$

$= 4 \left(\frac{x+1}{2} \right)$

$= 2x + 2$

7/ $V = \pi \int_{-3}^2 (y+3) dy$

$= \pi \left[\frac{y^2}{2} + 3y \right]_{-3}^2$ 1

$= \pi [(2+6) - (-4.5-9)]$

$= 12.5\pi \text{ units}^3$ 1

8/ 3

$\hat{A}XD = 90^\circ$ (angle in semi-c)

$\hat{A}XY = 90^\circ$ (adj. angles on str. line)

Similarly

$\hat{B}YC = 90^\circ$ (angle in semi-c)

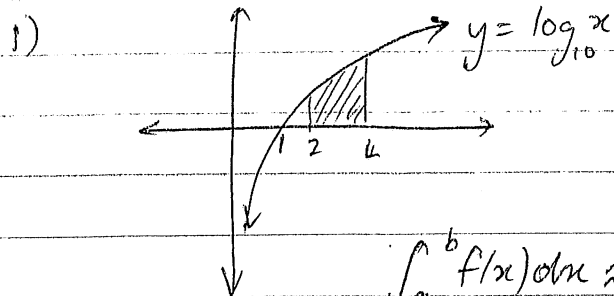
$\hat{B}YX = 90^\circ$ (adj. ang. on str. line)

$\Rightarrow AX \parallel BY$ since the co-interior angles AXY and BYX are supplementary.

$ABYX$ is a // gram with one angle a right angle.

is a rectangle.

Section B



$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

x	2	3	4
f(x)	$\log_{10} 2$	$\log_{10} 3$	$\log_{10} 4$

$$\int_2^4 \log_{10} x dx \approx \frac{4-2}{6} \left[\log_{10} 2 + 4 \log_{10} 3 + \log_{10} 4 \right]$$

$$\approx 0.94 \text{ units}^2$$

2) a) $f(x) = x^3 - 3x + 1$
 $f(0) = 1$
 $f(1) = -1$

since $f(x)$ is continuous and opposite in sign at endpoints, $f(x)$ has at least one zero when $0 \leq x \leq 1$.

b) $a_1 = a - \frac{f(a)}{f'(a)}$

$$f(x) = x^3 - 3x + 1$$

$$f(0) = 1$$

$$f'(x) = 3x^2 - 3$$

$$f'(0) = -3$$

$$a_1 = 0 - \frac{1}{-3}$$

$$a_1 = \frac{1}{3}$$

$$f\left(\frac{1}{3}\right) = \frac{1}{27}$$

$$a_2 = \frac{1}{3} - \frac{\left(\frac{1}{27}\right)}{\left(-\frac{8}{3}\right)}$$

$$f'\left(\frac{1}{3}\right) = -\frac{8}{3}$$

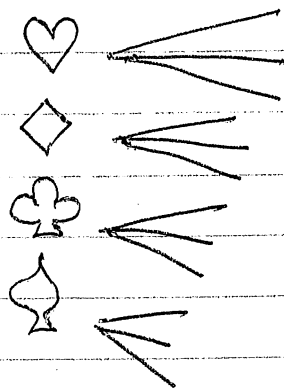
$$a_2 = \frac{25}{72}$$

$$a_2 \approx 0.347$$

c) $f'(1) = 0$ tangent when $x=1$ is horizontal
 $\therefore$ doesn't cross the x -axis.
 $\therefore$ Newton's method fails.

3) a) ${}^{13}C_3 \times {}^{13}C_2 = 22308$

b) 3 cards 2 cards



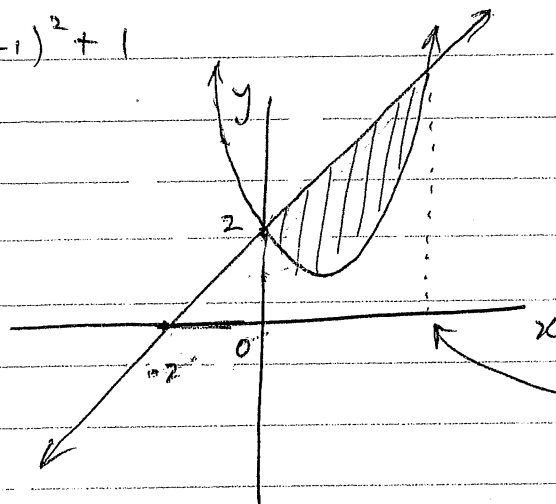
$12 \times 22308 = 267696$

4) $y = x^2 - 2x + 2$ — ①

$y = x + 2$ — ②

$y = x^2 - 2x + 1 + 1$

$y = (x-1)^2 + 1$



sub ① into ②

$x^2 - 2x + 2 = x + 2$

$x^2 - 3x = 0$

$x(x-3) = 0$

$x = 0$ $x = 3$

Area = $\int_0^3 [x+2 - (x^2 - 2x + 2)] dx$

$= \int_0^3 (3x - x^2) dx$

$= \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_0^3$

$$= \frac{3(3)^2}{2} - \frac{(3)^3}{3} - (0)$$

$$= \frac{9}{2} \text{ units}^2$$

$$5) {}^nP_k = 120 {}^nC_k$$

$$\frac{n!}{(n-k)!} = 120 \frac{n!}{(n-k)!k!}$$

$$k! = 120$$

$$\therefore k = 5$$

$$6) x^2 = 8y$$

$$x^2 = 4ay$$

$$\underline{a=2}$$

$$xx_1 = 2a(y+y_1)$$

$$xx_1 = 4(y+y_1)$$

equate

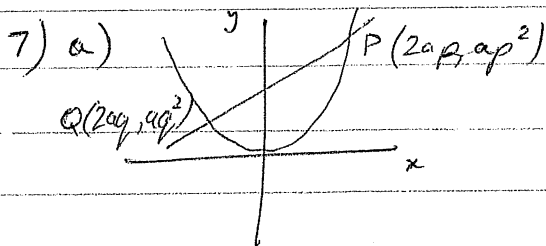
$$y = x-1$$

$$x = y+1$$

$$4x = 4(y+1)$$

$$x_1 = 4, y_1 = 1$$

∴ point of intersection of the tangents is (4, 1)



$$m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p^2 - q^2)}{2a(p - q)}$$

$$= \frac{(p-q)(p+q)}{2(p-q)}$$

$$= \frac{p+q}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = \frac{p+q}{2}(x - 2ap)$$

$$y - ap^2 = \frac{1}{2}(p+q)x - \cancel{ap^2} - apq$$

$$y - \frac{1}{2}(p+q)x + apq = 0$$

b) as $q \rightarrow p$

$$y - \frac{1}{2}(p+p)x + ap(p) = 0$$

$$y - px + ap^2 = 0$$

which is the equation of the tangent.

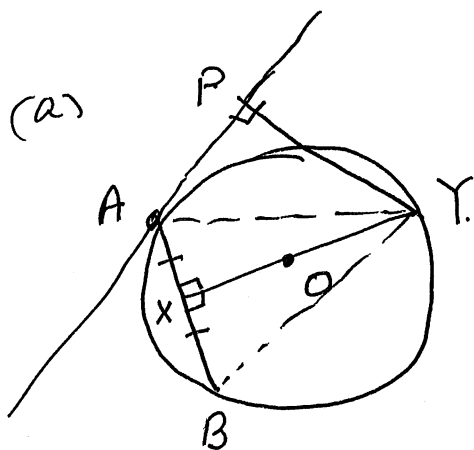
SECTION C. (EXT. I.)

L

Q1

$$\begin{aligned}
 \text{RHS} &= {}^{n-1}P_k + k {}^{n-1}P_{k-1} \\
 &= \frac{(n-1)!}{(n-1-k)!} + k \frac{(n-1)!}{(n-k)!} \\
 &= \frac{(n-1)!}{(n-k)!} [n-k + k] \\
 &= \frac{n(n-1)!}{(n-k)!} \\
 &= \frac{n!}{(n-k)!} \\
 &= {}^n P_r \\
 &= \text{LHS.}
 \end{aligned}$$

Q2.



(b)

$$\begin{aligned}
 AX &= BX \quad (\text{data}) \\
 \hat{A}XY &= \hat{B}XY \quad (\text{data}) \\
 XY &\text{ is common.}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \triangle AXY &\equiv \triangle BXY \quad (\text{S.A.S}) \\
 \therefore AY &= BY \quad (\text{corresponding sides of congruent triangles}) \\
 \therefore \triangle AYB &\text{ is isosceles.}
 \end{aligned}$$

(c) $\angle PAY = \alpha$.

$$\therefore \angle ABY = \alpha \quad (\text{angle between tangent and chord is equal to the angle in the alternate segment})$$

Now $\angle YAB = \angle ABY$ (isosceles triangle, base angles equal).

$$\therefore \angle YAB = \alpha.$$

(d) APYX is a cyclic quadrilateral (opposite angles $\widehat{APY}$ and $\widehat{AXY}$ are supplementary)

$\therefore \widehat{PYX} = \widehat{PY}$ (equal angles are subtended by equal chords, at the circumference of a circle).

[OR. Show that $\triangle AXY \equiv \triangle APY$ using (AAS) test]

Q3. $\frac{6!}{2! \times 2!} - \frac{5!}{2!} = 120$ OR. $\frac{4!}{2!} \times \binom{5}{2} = 120$.

Q4. $\frac{\binom{3}{2} \times \binom{17}{11}}{\binom{20}{13}} = \frac{91}{190}$.

Q5. Given $y = 2t^2$ & $x = 4t$.
 $\frac{dy}{dt} = 4t$ $\frac{dx}{dt} = 4$ $\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$
 $\frac{dt}{dx} = \frac{1}{4}$ $= 4t \times \frac{1}{4}$
 $= t$.

$\therefore$ slope of normal where $t = p$ is $-\frac{1}{p}$.

$\therefore$ eqn. of normal is $\frac{y - 2p^2}{x - 4p} = -\frac{1}{p}$.

$py - 2p^3 = -x + 4p$
 $x + py = 2p^3 + 4p$ — (1)

At $t = -p$, $\frac{y - 2p^2}{x + 4p} = \frac{1}{p} \therefore py - 2p^3 = x + 4p$
 $-x + py = 2p^3 + 4p$ — (2)

Add ① + ②

$$2py = 4p^3 + 8p$$

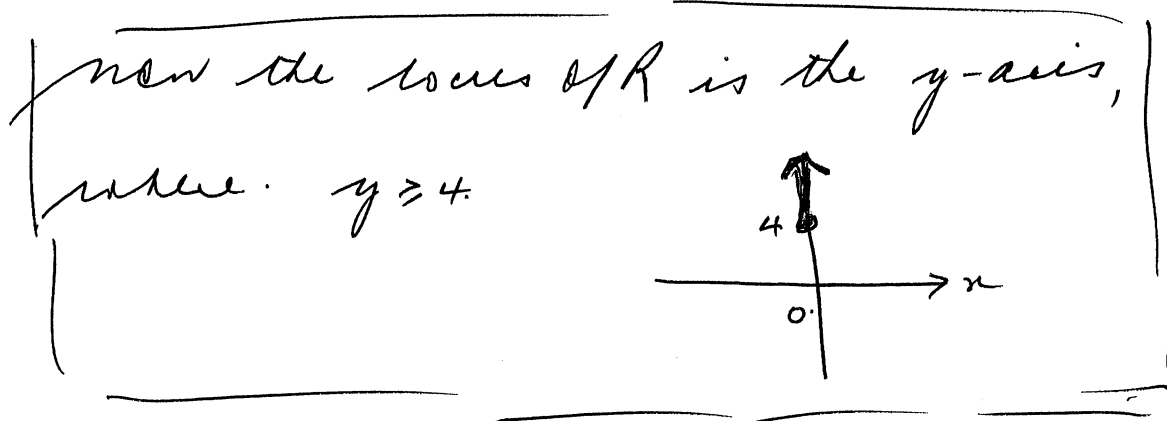
$$y = 2p^2 + 4$$

Subtract ② from ①.

$$2x = 0.$$

$$x = 0.$$

$\therefore R$ is $(0, 2p^2 + 4)$



Q6. Step I. Consider $n = 2$.

$$LHS = \frac{1}{2} + \frac{2}{3} = \frac{7}{6} \quad RHS = \frac{4}{3}$$

$\therefore$ true when $n = 2$ as $\frac{7}{6} < \frac{4}{3}$

Step II Assume the statement to be true when $n = k$.

$$\text{ie. } \frac{1}{2} + \frac{2}{3} + \dots + \frac{k}{k+1} < \frac{k^2}{k+1}.$$

Step III Required to prove. the statement is true
for $n = k+1$. under the assumption in Step II.

$$\text{ie. } \frac{1}{2} + \frac{2}{3} + \dots + \frac{k}{k+1} + \frac{k+1}{k+2} < \frac{(k+1)^2}{k+2}.$$

now

$$\begin{aligned}
 LHS &= \frac{1}{2} + \frac{2}{3} + \dots + \frac{k}{k+1} + \frac{k+1}{k+2} \\
 &< \frac{k^2}{k+1} + \frac{k+1}{k+2} \quad (\text{from assumption}) \\
 &= \frac{k^2(k+2) + (k+1)^2}{(k+1)(k+2)} \\
 &= \frac{k^3 + 2k^2 + k^2 + 2k + 1}{(k+1)(k+2)} \\
 &< \frac{k^3 + 3k^2 + 3k + 1}{(k+1)(k+2)} \\
 &= \frac{(k+1)^3}{(k+1)(k+2)} \\
 &= \frac{(k+1)^2}{k+2} = RHS
 \end{aligned}$$

$$\therefore \dots \frac{1}{2} + \frac{2}{3} + \dots + \frac{k+1}{k+2} < \frac{(k+1)^2}{k+2}$$

Step IV. By the Principle of mathematical Induction the statement is true for all integral $n \geq 2$.